

1. Encapsulation :

The Process of assuring to hide sensitive data and information from users.

Access Modifier is used for achieving Encapsulation and controlling the visibility & security level of each individual class and class members.

2. Inheritance :

The Process of inheriting fields and members from one class to another class.

There are 2 important categories in Inheritance.

Base Class (Parent Class) : The class that is inherited.

Derived Class (Child Class) : The class that inherits from another class.

3. Polymorphism :

The Process of using inherited classes' methods to perform a single action in different ways.

There are 2 important categories in Polymorphism.

Override (Runtime Polymorphism) : The Base Class's Method is being overridden by Derived Class's Method.

Overload (Compile time Polymorphism) : Methods with the same name but different parameters.

4. Abstraction :

The Process of hiding certain details and showing only essential information to the users.

Abstraction can be achieved with either Abstract Classes or Interfaces.

Abstract Class : It is a restricted class that can not be used to create objects. To access abstract class, it must be inherited from another class. An Abstract Class can have both Abstract Method and Regular Method. Abstract Method does not have a body. The body is provided by the Derived Class (inherited from the Base Class).

Interface : Another way to achieve abstraction in c#, is with interfaces. An interface is like a completely "Abstract Class" which can only contain Abstract Methods and Properties. To access the Interface Methods, the Interface must be implemented by another class. The body of Interface Method is provided by the Implement Class. Like Abstract Class, Interface can not be used to create object. C# does not support Multiple Inheritance but it can be achieved with Interfaces. A class can implement multiple interfaces.